Univariate Exploratory Data Analysis

Overview

Univariate Exploratory Data Analysis (EDA) was conducted to examine the distribution and characteristics of individual variables within the Netflix dataset. The primary objective of this phase was to understand the structure of the dataset by analyzing each important feature independently before investigating relationships between multiple variables.

The analysis focused on four investigation questions related to content categories, genres, release trends, and geographical distribution. Various statistical summaries and visualizations were used to identify dominant patterns, frequency distributions, and temporal trends within the dataset.

Methodology

The univariate analysis followed a structured workflow to ensure consistency and reproducibility across all investigation questions.

The key steps performed during this phase are outlined below:

1. Loaded the cleaned Netflix dataset into a Pandas DataFrame.
2. Selected the variables required for each investigation question.
3. Calculated frequency distributions using the value_counts() function.
4. Computed percentage distributions to compare the relative contribution of different categories.
5. Processed multi-valued categorical variables (Type and Country) by:
    - Splitting comma-separated values.
    - Expanding them into individual observations using the explode() function.
    - Removing unnecessary whitespace.
6. Converted the Release_Date column into datetime format.
7. Extracted the release year for temporal trend analysis.
8. Created summary tables containing counts and percentage distributions.
9. Visualized the results using:
    - Bar Charts
    - Horizontal Bar Charts
    - Pie Charts
    - Combined Bar and Line Charts
10. Interpreted each visualization through data-driven insights, business interpretation, and conclusions.

Investigation Question 1

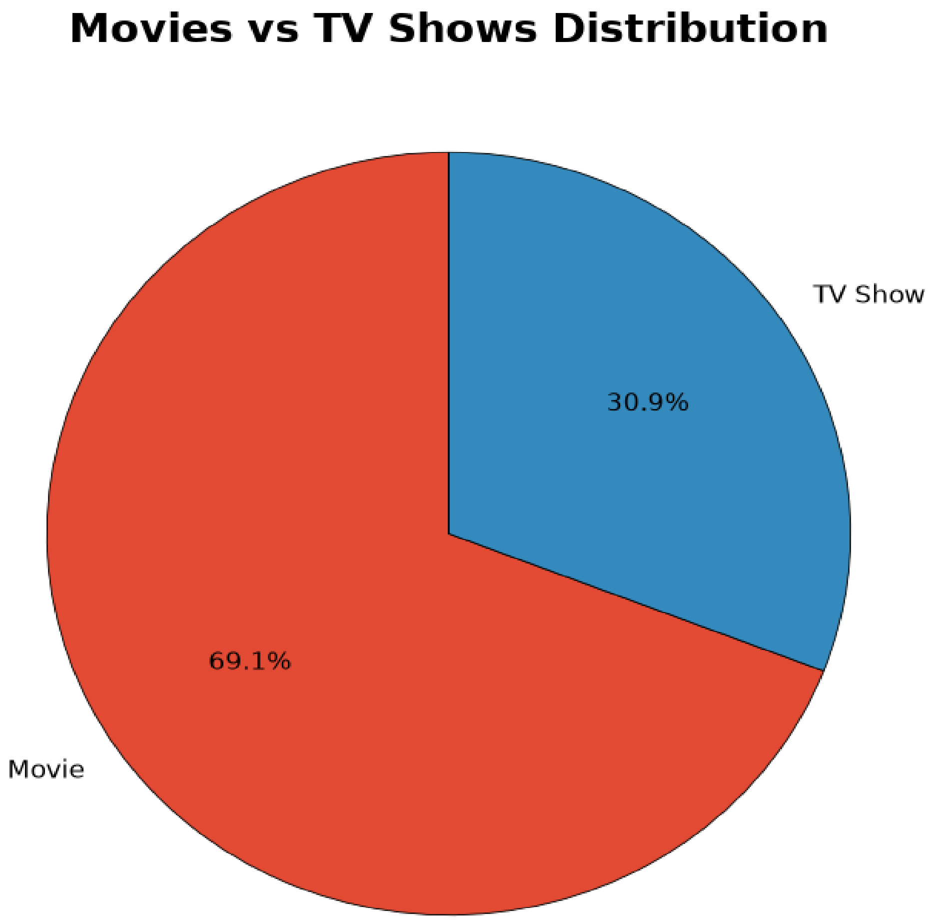
What is the distribution of Movies versus TV Shows in Netflix's catalog?

Objective

To determine the proportion of Movies and TV Shows available on Netflix and understand the platform's overall content composition.

Analysis Performed

- Calculated the frequency of each content category.
- Computed percentage distribution.
- Created a summary table.
- Visualized the distribution using both a bar chart and a pie chart.



Findings

- Movies account for **69.14%** of the Netflix catalog.
- TV Shows account for **30.86%** of the catalog.
- Movies outnumber TV Shows by approximately **2.2 to 1**.

Interpretation

The analysis indicates that Netflix maintains a movie-focused catalog while still offering a substantial collection of television series to encourage long-term viewer engagement.

Investigation Question 2

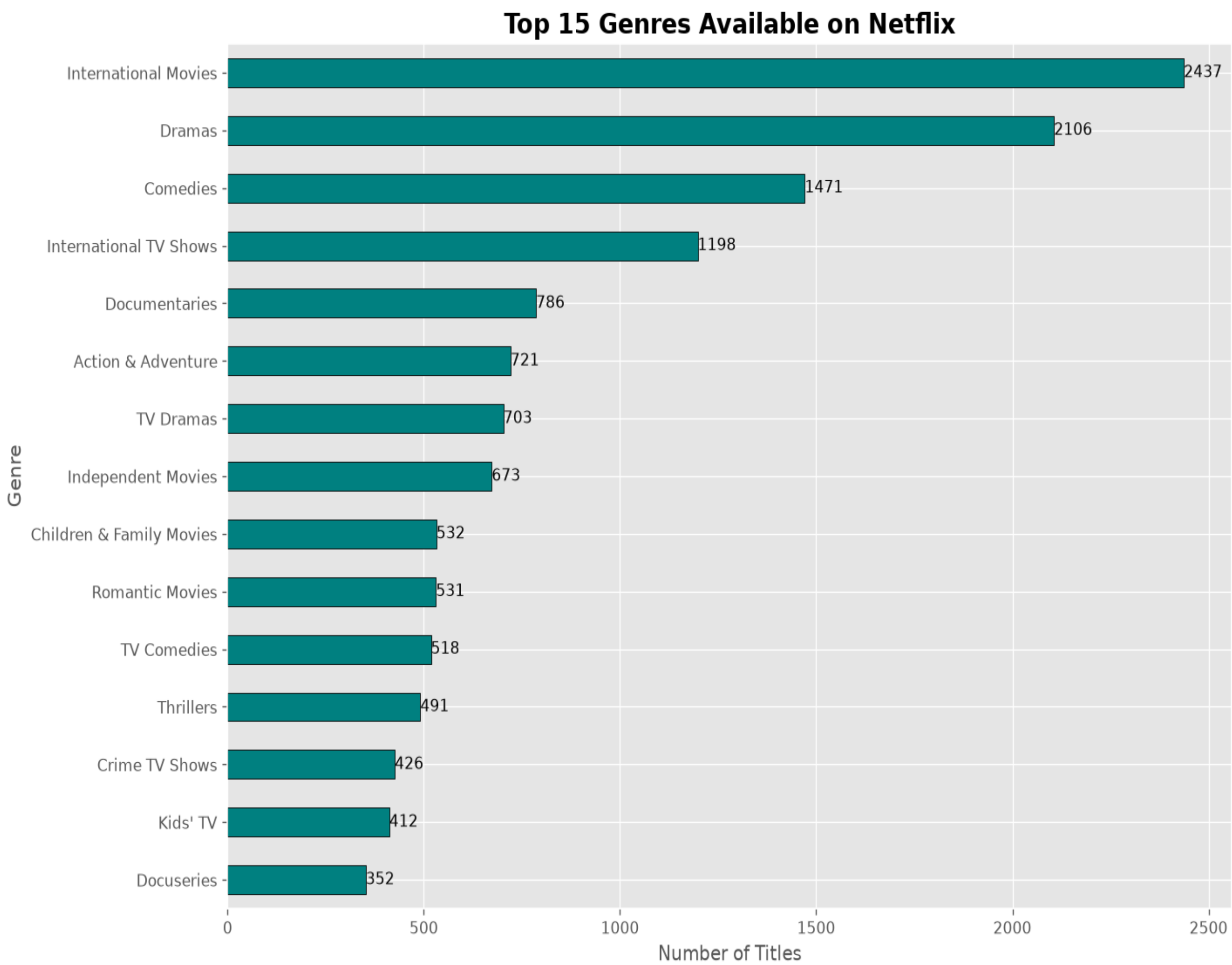
Which genres or content types are most prevalent in Netflix's library?

Objective

To identify the most frequently occurring genres available on Netflix.

Analysis Performed

- Split the multi-valued Type column.
- Expanded individual genres using the explode() function.
- Calculated frequency and percentage distributions.
- Ranked genres based on occurrence.
- Visualized the top genres using a horizontal bar chart.



Findings

The most common genres identified were:

Rank	Genre	Count	Percentage
1	International Movies	2,437	14.29%
2	Dramas	2,106	12.35%
3	Comedies	1,471	8.63%
4	International TV Shows	1,198	7.03%
5	Documentaries	786	4.61%

Interpretation

Netflix emphasizes internationally produced content while maintaining a balanced catalog of dramas, comedies, documentaries, action titles, children's content, and romantic programming. This diversified genre portfolio enables the platform to appeal to audiences with varying entertainment preferences.

Investigation Question 3

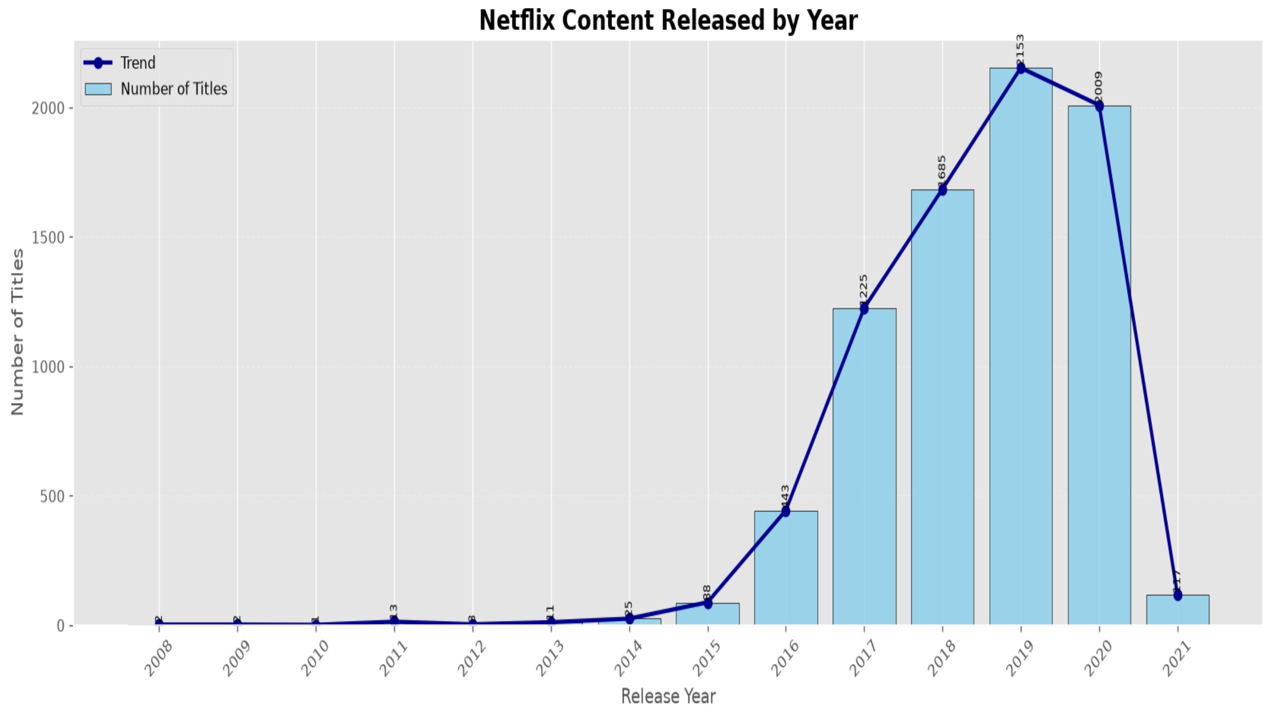
How has Netflix's content volume changed over time?

Objective

To analyze how Netflix's content library has expanded over the years.

Analysis Performed

- Converted the Release_Date column into datetime format.
- Extracted the release year.
- Grouped the dataset by year.
- Counted titles released each year.
- Visualized the results using a combined bar and line chart.



Findings

Year	Titles Released
2012	3
2013	11
2014	25
2015	88
2016	443
2017	1,225
2018	1,685
2019	**2,153**
2020	2,009
2021	117

Interpretation

Netflix experienced rapid content expansion between **2015 and 2019**, reaching its highest annual release volume in **2019**. The slight decline in 2020 was followed by a sharp reduction in 2021, which is likely influenced by the dataset's collection period rather than an actual decrease in Netflix's production strategy.

Investigation Question 4

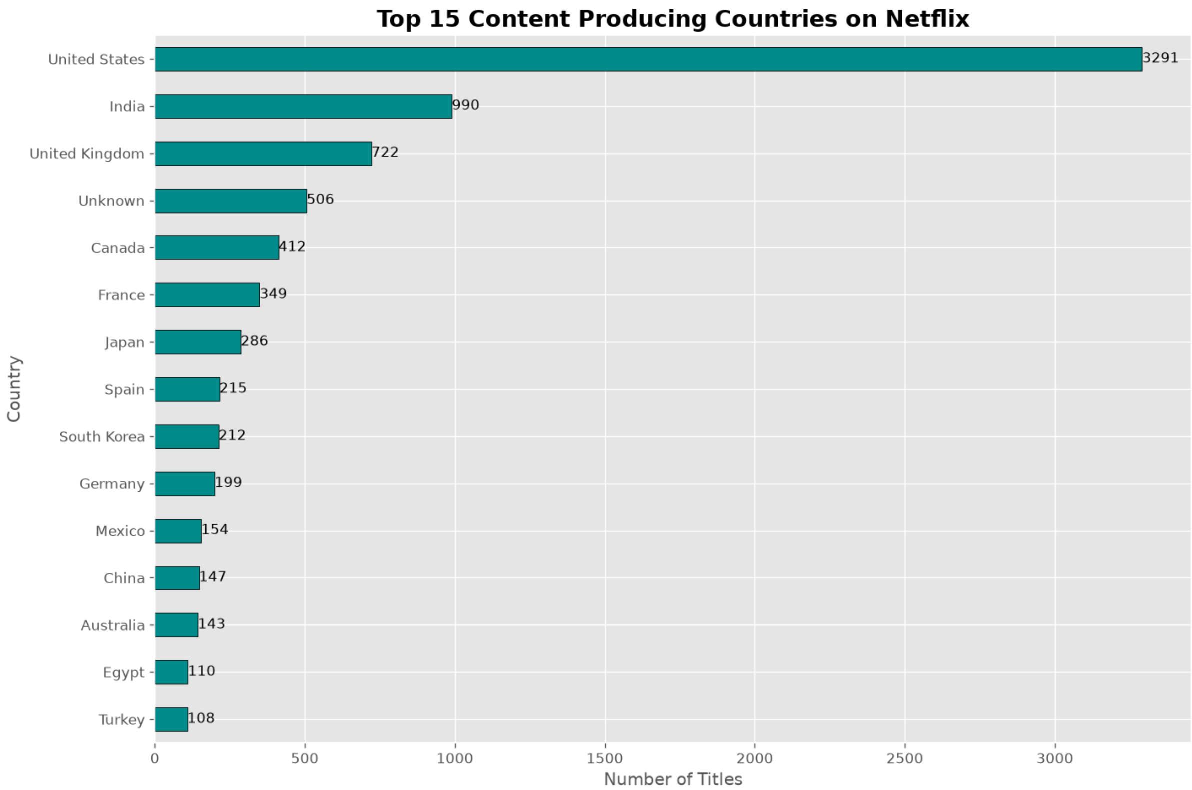
Which countries produce the most content available on Netflix?

Objective

To identify the countries contributing the largest number of titles to Netflix's content library.

Analysis Performed

- Split the multi-country entries within the Country column.
- Expanded the values into separate observations.
- Calculated frequency and percentage distributions.
- Ranked countries according to content contribution.
- Visualized the leading countries using a horizontal bar chart.



Findings

Rank	Country	Count	Percentage
1	United States	3,291	34.41%
2	India	990	10.35%
3	United Kingdom	722	7.55%
4	Unknown	506	5.29%

Rank	Country	Count	Percentage
5	Canada	412	4.31%

Interpretation

The United States dominates Netflix's content library, contributing over one-third of all titles. India and the United Kingdom are also major contributors, highlighting Netflix's strong presence in key entertainment markets. The inclusion of content from numerous other countries demonstrates the platform's commitment to maintaining a globally diverse catalog.

Overall Findings

The univariate analysis revealed several important characteristics of Netflix's content library:

- Movies constitute approximately **69%** of the catalog, indicating that Netflix primarily focuses on feature-length content.
- International Movies, Dramas, and Comedies are the platform's most prevalent genres, reflecting a strategy centered on globally appealing entertainment.
- Netflix experienced substantial growth in content additions between **2015 and 2019**, with **2019** recording the highest number of releases.
- The United States remains Netflix's primary content source, followed by India and the United Kingdom, illustrating the platform's strong presence in major production markets.
- The presence of internationally produced content across multiple genres demonstrates Netflix's investment in serving a diverse global audience.

Key Business Insights

The analysis suggests that Netflix has adopted a globally diversified content strategy that balances mainstream entertainment with region-specific productions. The dominance of internationally produced movies highlights the company's focus on expanding localized content while maintaining worldwide accessibility.

The rapid increase in content releases between 2015 and 2019 reflects Netflix's aggressive investment in original productions, licensing agreements, and international market expansion. Although movies dominate the platform's catalog, the continued presence of television series supports long-term subscriber engagement through serialized storytelling.

Overall, the findings indicate that Netflix has successfully developed a comprehensive and geographically diverse content library designed to satisfy a broad range of audience preferences across different regions and cultures.

Conclusion

The Univariate Exploratory Data Analysis provided a detailed understanding of the individual characteristics of the Netflix dataset by examining content categories, genres, release trends, and country-wise distribution independently. This phase established a solid analytical foundation by identifying the dominant patterns and overall composition of the dataset.

The insights gained during this stage will serve as the basis for the subsequent **Bivariate Exploratory Data Analysis**, where relationships between variables—such as ratings and genres, categories and countries, and other paired attributes—will be explored to uncover deeper patterns and interactions within the Netflix content library.